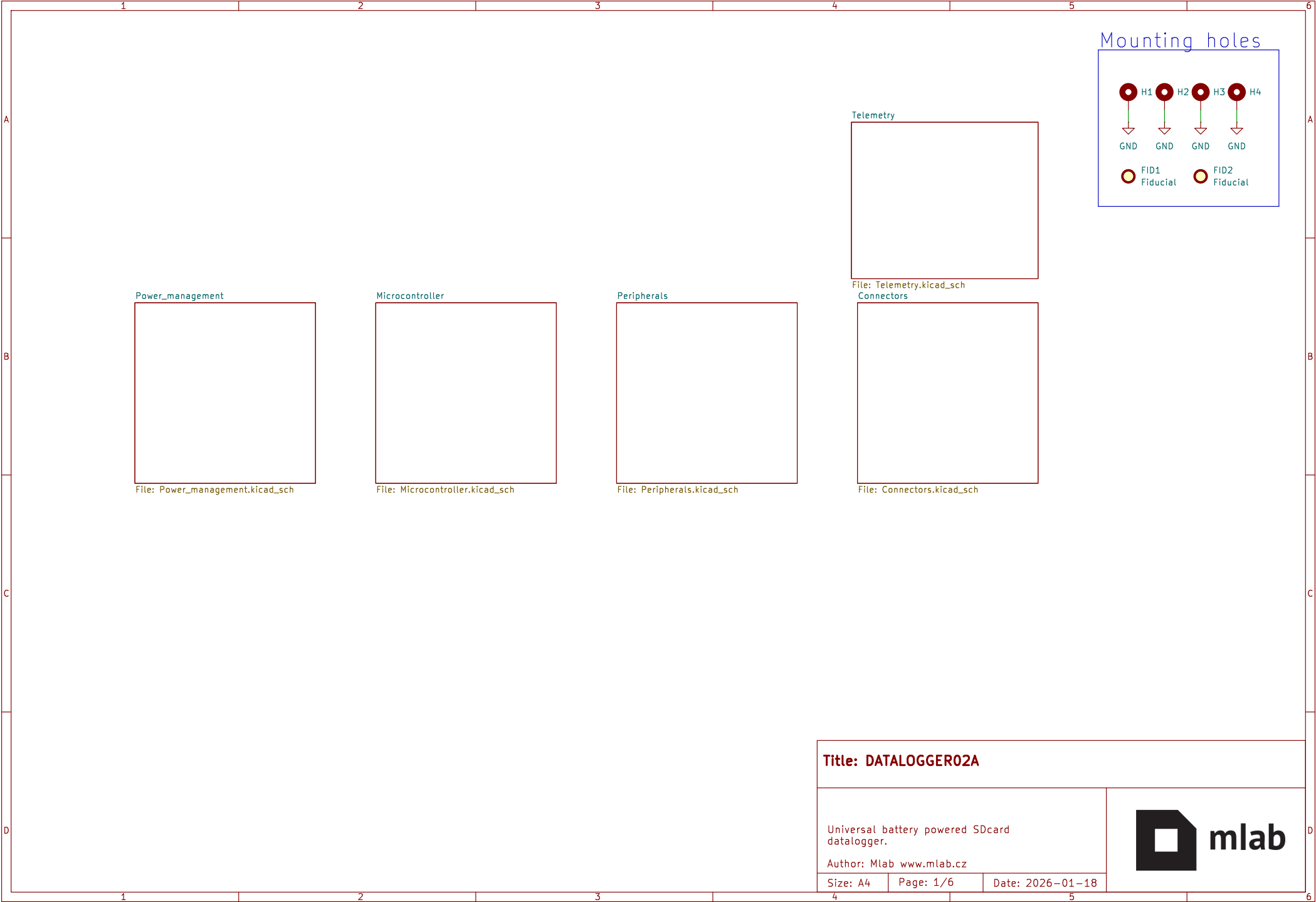
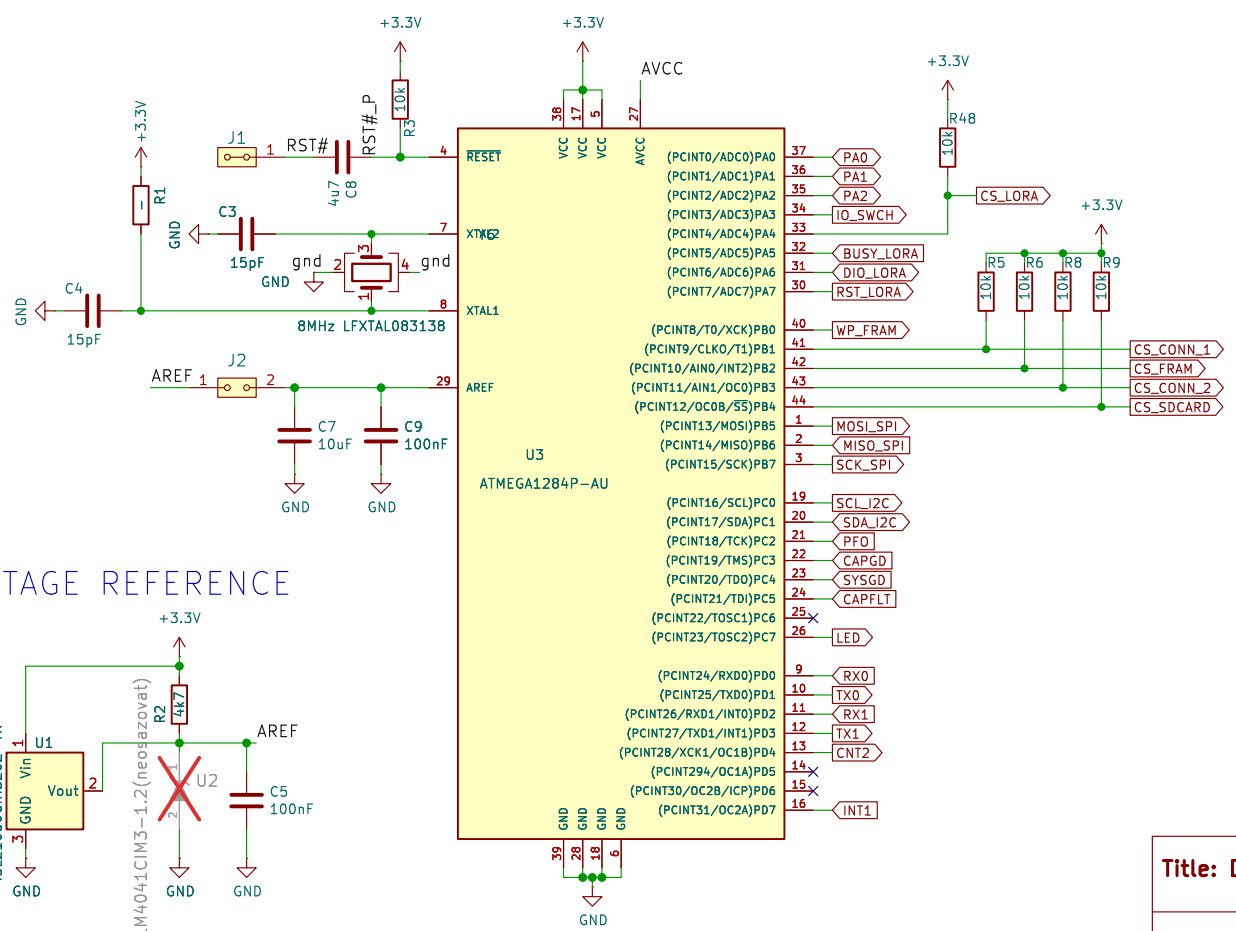
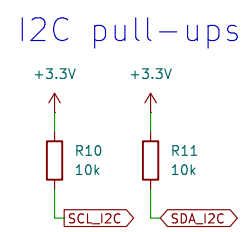
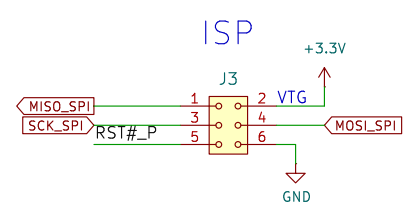
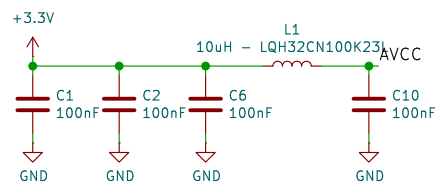
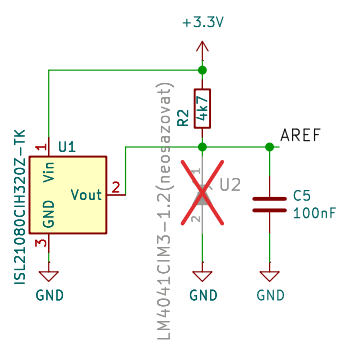






- PB0 - CS_CONN_1
- PB1 - CS_FRAM
- PB2 - CS_CONN_2
- PB4 - CS_SDCARD
- PB5 - MOSI_SPI
- PB6 - MISO_SPI
- PB7 - SCK_SPI

VOLTAGE REFERENCE



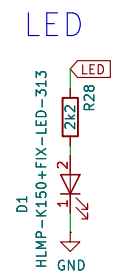
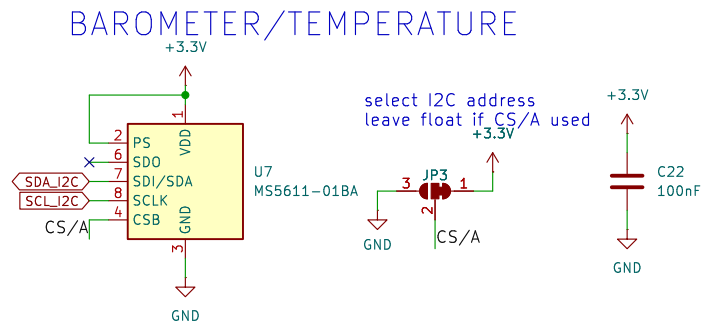
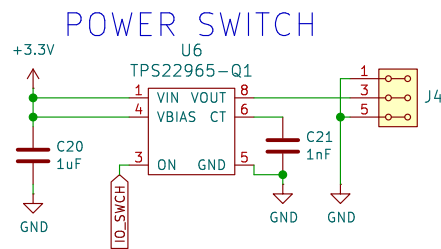
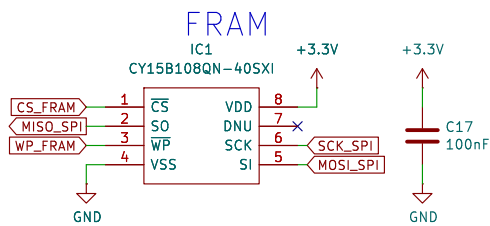
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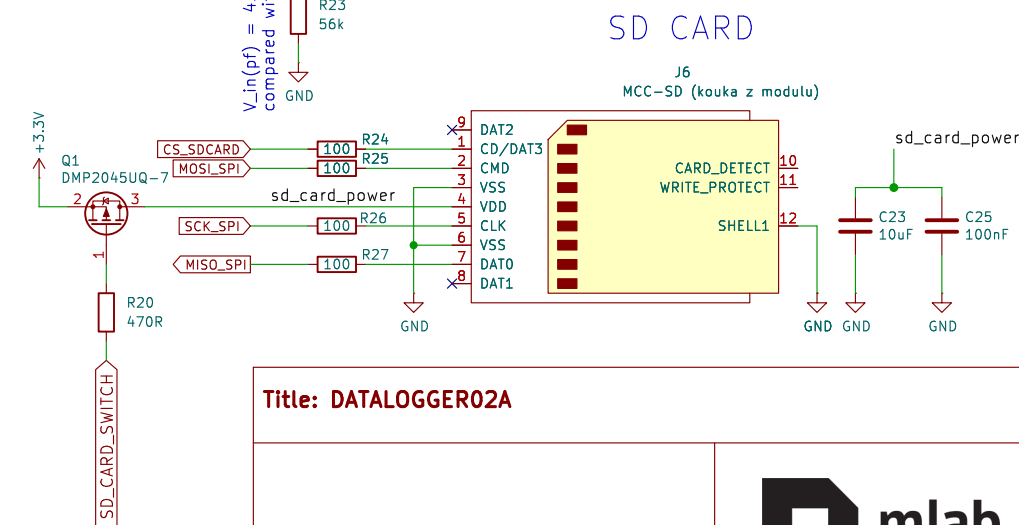
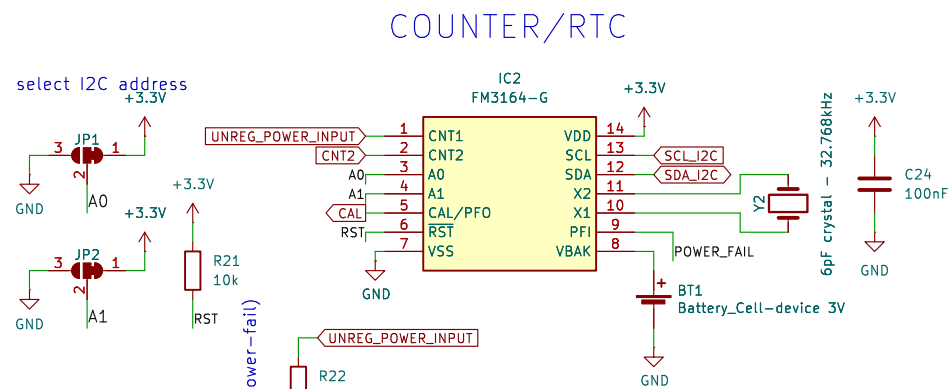
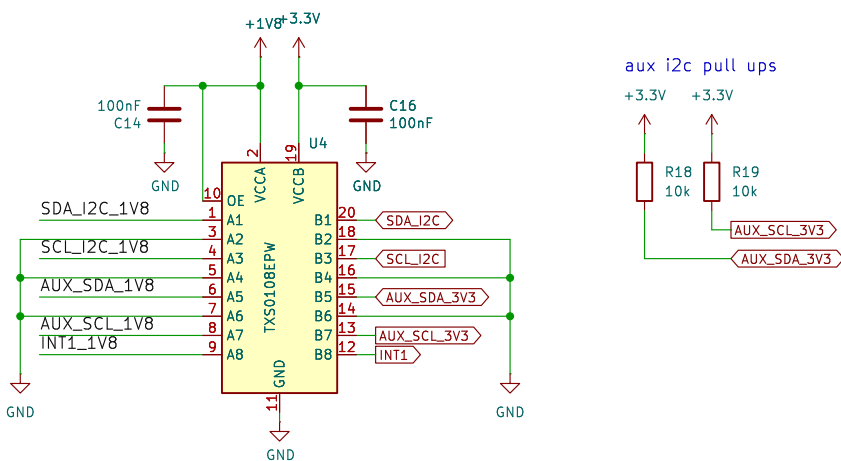
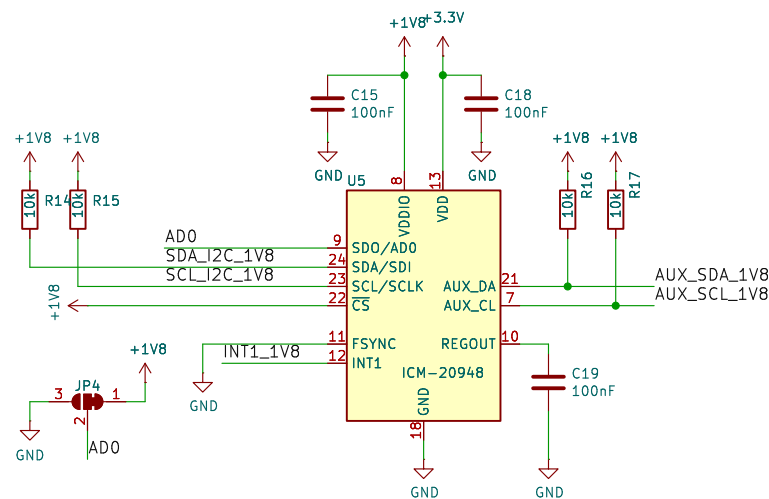
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INERTIAL MEASUREMENT UNIT



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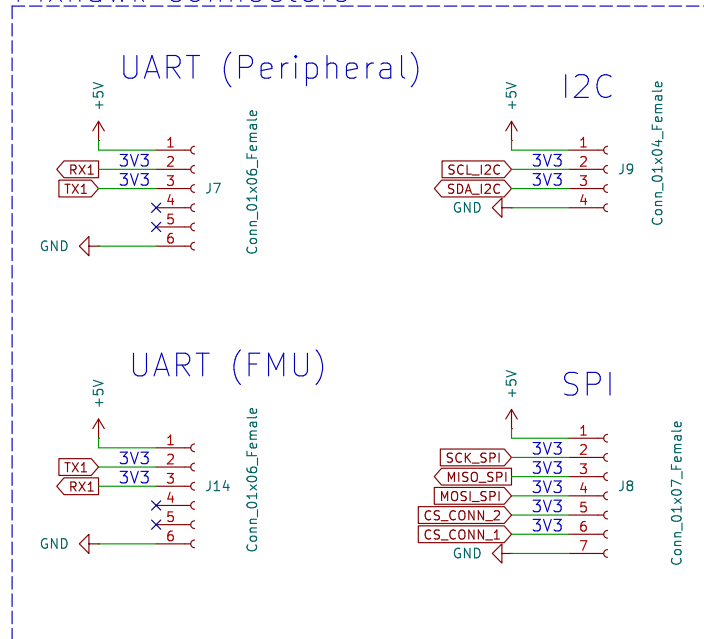
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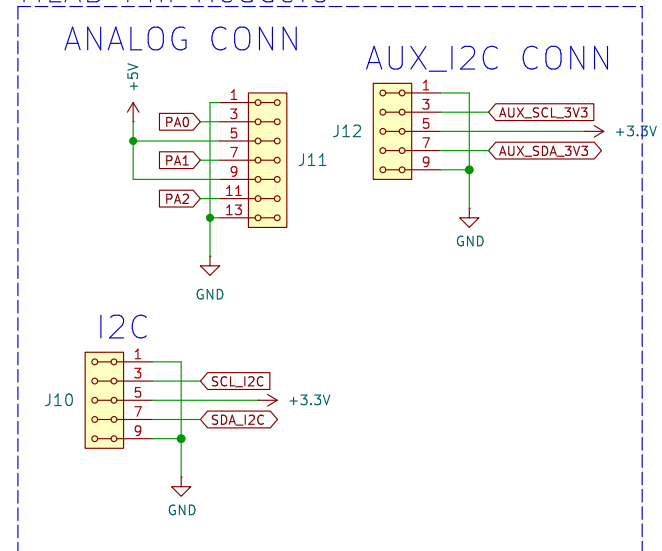
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Pixhawk connectors



MLAB Pin Headers



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SUPERCAP MANAGEMENT

Supercap charge current = 0 A
if load current = 1 A
 $R_s = 25 \text{ mV} / I_{\text{load}}$

BUCK-BOOST CONVERTOR
1V6 – 5V5 IN
3V3 OUT

BUCK-BOOST CONVERTOR
1V6 – 5V5 IN
5V OUT

LDO
3V3 IN
1V8 OUT

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mlab

BUCK-BOOST CONVERTOR
1V6 – 5V5 IN
3V3 OUT

BUCK-BOOST CONVERTOR
1V6 – 5V5 IN
5V OUT

LDO
3V3 IN
1V8 OUT

Supercap Management
Supercap charge current = 0 A
if load current = 1 A
 $R_s = 25 \text{ mV} / I_{\text{load}}$

Na IMON lze merit I_{load}, ale musi tam byt follower

V_{in}(pf) = 4.4 V (power-fail)

V_{load} = 4.984 V

V_{chg} = 2.56 V (charge voltage)

L_{chg} = 1 A (charge current)

UNREG_POWER_INPUT

IC3 LTC4041EUFDPB

U10 TPS631000DRLR

U9 TPS631000DRLR

U8 MIC5504-1.8YM5-TR

JP5 ModeSelect

JP6 ModeSelect

JP7 ModeSelect

JP8 ModeSelect

JP9 ModeSelect

JP10 ModeSelect

JP11 ModeSelect

JP12 ModeSelect

JP13 ModeSelect

JP14 ModeSelect

JP15 ModeSelect

JP16 ModeSelect

JP17 ModeSelect

JP18 ModeSelect

JP19 ModeSelect

JP20 ModeSelect

JP21 ModeSelect

JP22 ModeSelect

JP23 ModeSelect

JP24 ModeSelect

JP25 ModeSelect

JP26 ModeSelect

JP27 ModeSelect

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JP217 ModeSelect

JP218 ModeSelect

JP219 ModeSelect

JP220 ModeSelect

JP221 ModeSelect

JP222 ModeSelect

JP223 ModeSelect

JP224 ModeSelect

JP225 ModeSelect

JP226 ModeSelect

JP227 ModeSelect

JP228 ModeSelect

JP229 ModeSelect

JP230 ModeSelect

JP231 ModeSelect

JP232 ModeSelect

JP233 ModeSelect

JP234 ModeSelect

JP235 ModeSelect

JP236 ModeSelect

JP237 Mode

BUCK-BOOST CONVERTOR
1V6 – 5V5 IN
3V3 OUT

BUCK-BOOST CONVERTOR
1V6 – 5V5 IN
5V OUT

LDO
3V3 IN
1V8 OUT

Supercap Management
Supercap charge current = 0 A
if load current = 1 A
 $R_s = 25 \text{ mV} / I_{\text{load}}$

Component Labels and Values:
J24: HEADER_2x03_PARALLEL
F1: 1.2A
DMTH3002LPS: Q2, Q3
R37: 25mR
R35: 6.2k 1/8W
IC3: LTC4041EUFDPBFB
C39: 22uF
C34: 47uF
C29: 47uF
C32: 100nF
R38: 523k (E96)
R41: 100k
R40: 220k
XAL-6030-182: L2
C28: 50F Supercap
R39: 100k
R30: 56k
R29: 150k
R31: 1M
R32: 100k
R33: 100k
R34: 100k
R42: 806k
R43: 91k
R44: 511k
R45: 91k
C13: 47uF
C33: 22uF
C12: 47uF
C11: 22uF
C27: 1uF
C31: 1uF
C26: 1nF
C25: 1nF
C24: 1nF
C23: 1nF
C22: 1nF
C21: 1nF
C20: 1nF
C19: 1nF
C18: 1nF
C17: 1nF
C16: 1nF
C15: 1nF
C14: 1nF
C13: 1nF
C12: 1nF
C11: 1nF
C10: 1nF
C9: 1nF
C8: 1nF
C7: 1nF
C6: 1nF
C5: 1nF
C4: 1nF
C3: 1nF
C2: 1nF
C1: 1nF

Pin Headers:
J5: HEADER_2x03_PARALLEL
JP5: ModeSelect
JP6: ModeSelect

IC Pin Connections:
IC3: LTC4041EUFDPBFB
IC4: MIC5504-1.8YM5-TR
IC5: TPS631000DRLR
IC6: TPS631000DRLR

Other Labels:
UNREG_POWER_INPUT
V_in(pf) = 4.4 V (power-fail)
V_load = 4.984 V
V_chg = 2.56 V (charge voltage)
L_chg = 1 A (charge current)
Na IMON lze merit I_load, ale musi tam byt follower

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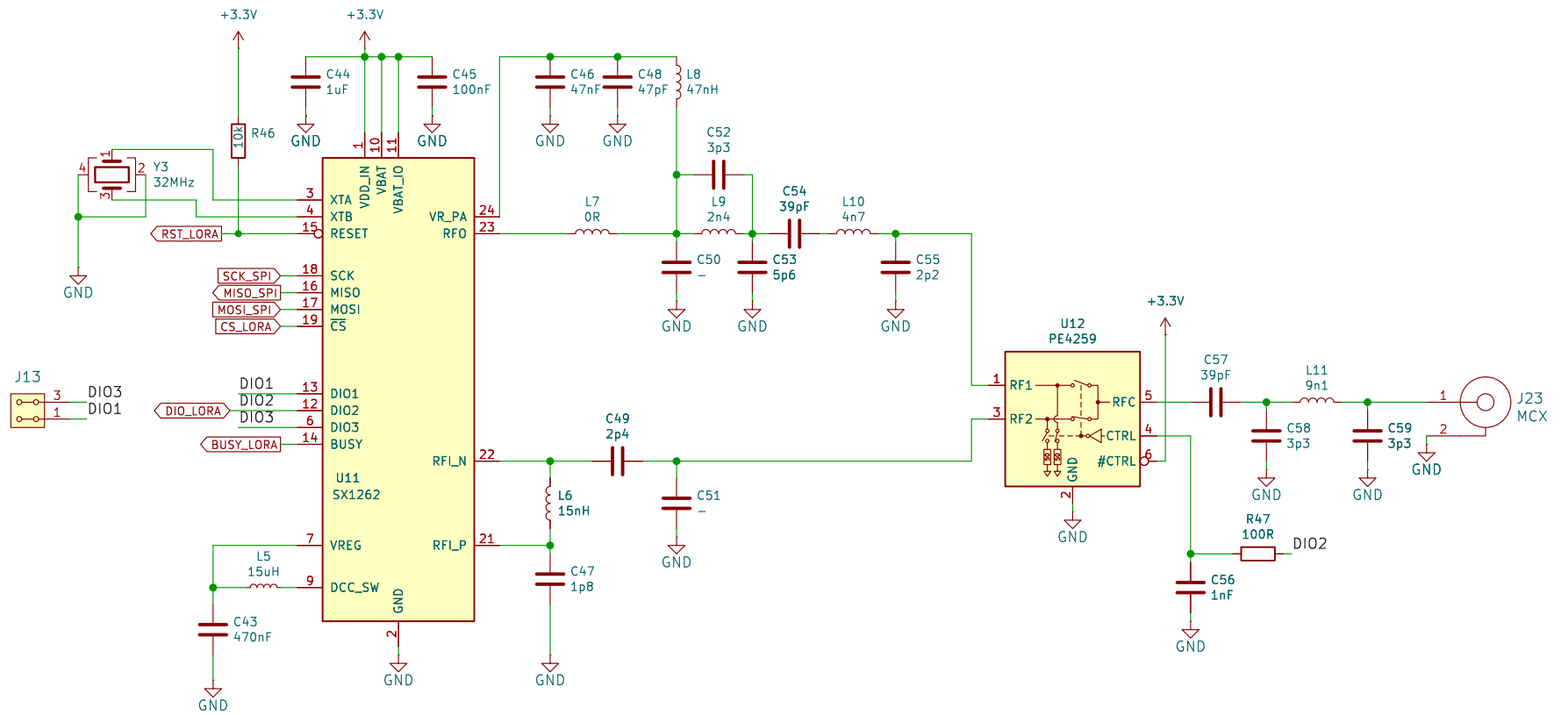
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